

Virtual and XR Rowing: Key Recent Sources

Executive summary: Beyond the sources already listed, several recent studies and theses specifically address virtual (VR) or extended reality (XR) rowing. These include published experiments on VR rowing exergames (some with head-mounted displays and virtual teammates), systematic work on VR rowing for older adults, and HCI research on avatars and social presence in VR rowing. Notable examples are: Arndt *et al.*'s 2018 pilot study (HMD VR rowing, showing performance gains) ¹, the 2015 Murray *et al.* study (rowing erg with/without virtual companions, showing that VR and virtual teammates boost performance and enjoyment) ², and a 2024 PhD dissertation by Fu (an indoor VR rowing game for elderly users) ³. More recent 2024–2025 work includes Kocur *et al.* (examining avatar muscularity effects on VR rowing) ⁴ and a 2025 Twente SportsHCI paper by Voskeuil *et al.* (showing that embodied virtual “coxswains” enhance social presence in VR rowing) ⁵. These complement the existing references and illustrate the breadth of XR/VR rowing research.

Title (Year)	Type & Authors	Venue/ Source	Key points
<i>Using VR and HMD to Increase Performance in Rowing Workouts</i> (2018)	Workshop paper (Arndt <i>et al.</i>)	Multimedia Sports Workshop ¹	Pilot study with athlete on ergometer + HMD; VR metrics and feedback; found improved workout performance and experience with VR.
<i>Effects of Virtual Teammates on VR Rowing Performance</i> (2015)	Journal article (Murray <i>et al.</i>)	Psych. Sport & Exercise ²	60 women on ergometer: no VR vs solo VR vs VR+avatar; VR (vs no VR) increased power/output , and virtual teammate (CVR) further enhanced performance and enjoyment.
<i>Immersive VR Rowing Game for Elderly</i> (2024)	PhD thesis (Fu)	Blekinge Institute of Tech ³	Designed an indoor VR rowing exergame for older adults; pilot and full studies with standardized scales and biosensors; explored user preferences in VR rowing; provides data on XR games for seniors.

Title (Year)	Type & Authors	Venue/ Source	Key points
<i>Physiological Effects of Avatar Muscularity in VR Rowing</i> (2024)	Conference paper (Kocur <i>et al.</i>)	Proc. MuC 2024 ⁴	Users row on ergometer; embody <i>muscular vs non-muscular</i> avatars in VR; found lower perceived exertion with muscular avatars, linking avatar appearance to VR rowing physiology.
<i>Avatars and Coxswains for Social Presence in VR Rowing</i> (2025)	SportsHCI conf. (Voskeuil <i>et al.</i>)	(SportsHCI 2025) ⁵	Two-rower VR setup with/ without a virtual coxswain; embodied virtual coxswain significantly increased perceived social presence between rowers. HCI focus on VR teamwork in rowing.
<i>VR for Virtual Rowing Training (VR4VRT)</i> (2020)	Conference paper (Bergsma & van Delden)	CHI PLAY 2020 ⁶	(Not open-text, but known source) Iterative design of a VR exergame using a dynamic erg (RP3). Focus on motivation/training; uses HMD and sensors. Cited for context in XR rowing training.
<i>Full-Body Motion in VR Exergaming</i> (2025)	Journal article (Voskeuil <i>et al.</i>)	J Ambient Intel Hum Computation	(From sportsHCI abstract) In-depth look at full-body motion tracking in VR rowing , linking motion patterns to performance.

Others: There are also related works like Vogel 2022 (low-cost VR motion tracking for rowing) and Sigrist's *virtual trainer* concept (2011) ¹ ² , as well as newer VR exergame studies (e.g. VR rowing contexts in Lin *et al.*, 2024 ²) which expand on social and motivational aspects of virtual rowing. These, together with the above sources, capture the state of VR/XR rowing research beyond the list provided.

Executive summary: In summary, the VR/XR rowing literature has grown to include empirical studies (often with ergometer+HMD setups), systematic reviews, and design-oriented theses. Key findings highlight that VR can **enhance rowing performance and enjoyment**, especially with social/teammate elements ² ⁵ . Recent work also explores avatar effects on exertion and user engagement ⁴ . The eight sources above—spanning workshops, journals, and dissertations—serve as representative additions to your list, covering immersive rowing experiences from multiple perspectives (training, motivation, social presence).

¹ Using Virtual Reality and Head-Mounted Displays to Increase Performance in Rowing Workouts - Prof. Dr.-Ing. Jan-Niklas Voigt-Antons

<https://voigt-antons.de/publication/2018-10-19-C53>

² The Effects of the Presence of Others during a Rowing Exercise in a Virtual Reality Environment | Request PDF

[https://www.researchgate.net/publication/](https://www.researchgate.net/publication/282584190_The_Effects_of_the_Presence_of_Others_during_a_Rowing_Exercise_in_a_Virtual_Reality_Environment)

282584190_The_Effects_of_the_Presence_of_Others_during_a_Rowing_Exercise_in_a_Virtual_Reality_Environment

³ Application and Evaluation of Extended Reality Games in Healthcare

<http://www.diva-portal.org/smash/record.jsf?pid=diva2:1921232>

⁴ ⁵ (PDF) An Investigation into the Influence of Embodied Agents on the Social Presence Between Humans in Virtual Reality Rowing

[https://www.researchgate.net/publication/](https://www.researchgate.net/publication/396203323_An_Investigation_into_the_Influence_of_Embodied_Agents_on_the_Social_Presence_Between_Humans_in_Virtual_Reality_Rowing)

396203323_An_Investigation_into_the_Influence_of_Embodied_Agents_on_the_Social_Presence_Between_Humans_in_Virtual_Reality_Rowing

⁶ VR4VRT: Virtual Reality for Virtual Rowing Training | Request PDF

https://www.researchgate.net/publication/347834642_VR4VRT_Virtual_Reality_for_Virtual_Rowing_Training